Single-cell RNA sequencing reveals tumor progression and microenvironment features in immune-hot and immune-cold of hepatocellular carcinoma

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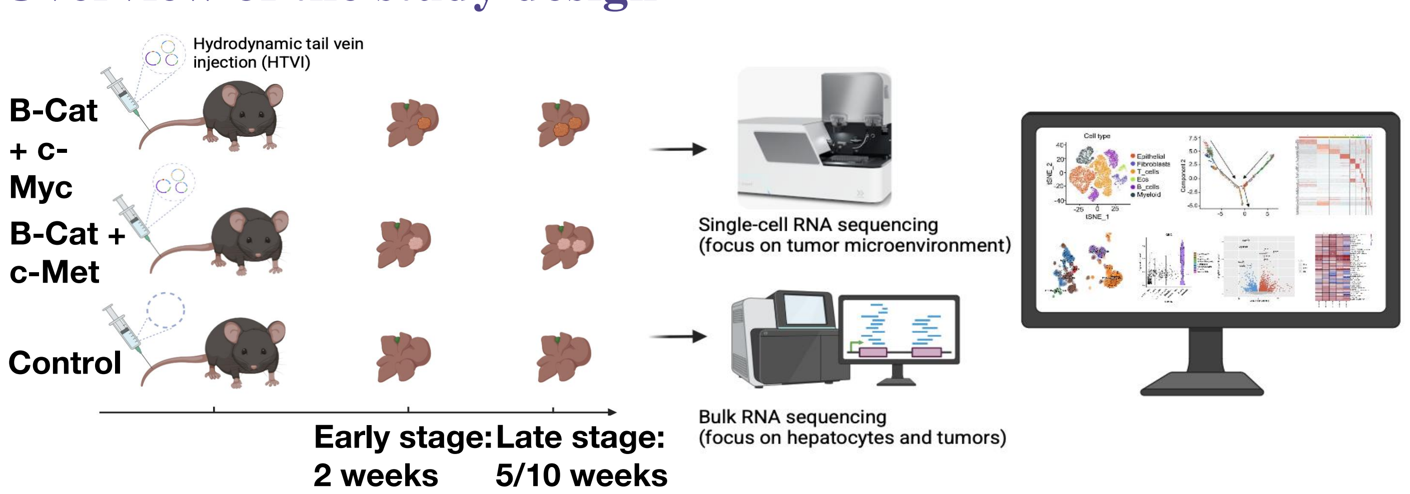
Background

Hepatocellular carcinoma (HCC) has tripled in incidence in the United States over the past 40 years, with more than 40,000 new cases diagnosed annually. While HCC can be curatively resected in the early stages, its high recurrence rate and lack of effective treatments result in a 5-year survival rate of only 18%. Our previous multi-omics analysis of Hispanic HCC identified two molecular subtypes with distinct immune activity and prognosis, which refer to immune hot (HM-1) and immune cold tumor (HM-2). However, the mechanisms that drive and shape these differences in the tumor and its microenvironment remain unclear.

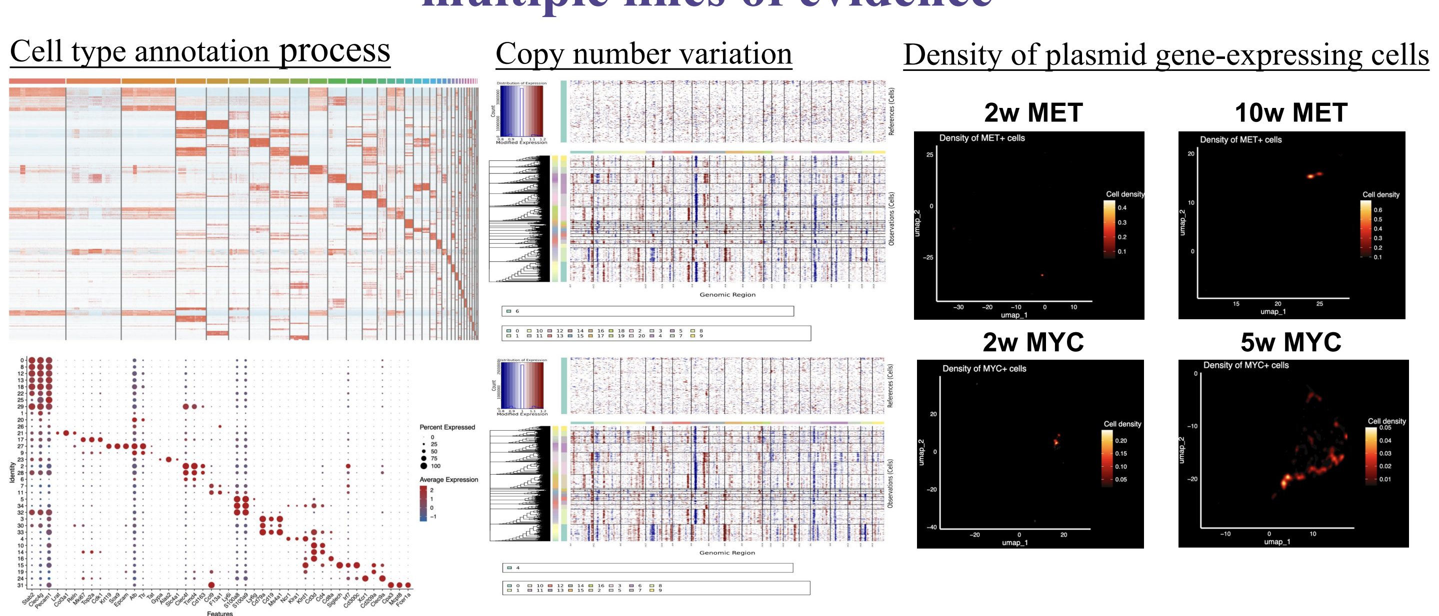
Methods

We generated spontaneous HCC mouse models using hydrodynamic tail vein injection of Sleeping Beauty transposon plasmids encoding human β-catenin combined with either MYC or MET. We observed that these two HCC mouse models exhibited phenotypes similar to HM-1 and HM-2, respectively. Single-cell RNA sequencing was performed on liver tissue (tumor microenvironment) at 2, 5, and 10 weeks post-injection, profiling 61,734 high-quality cells to capture temporal and cell type-specific transcriptional changes. In parallel, bulk RNA-seq from the liver tissues was used to characterize tumor-intrinsic molecular features in MYC- and MET-driven tumors.

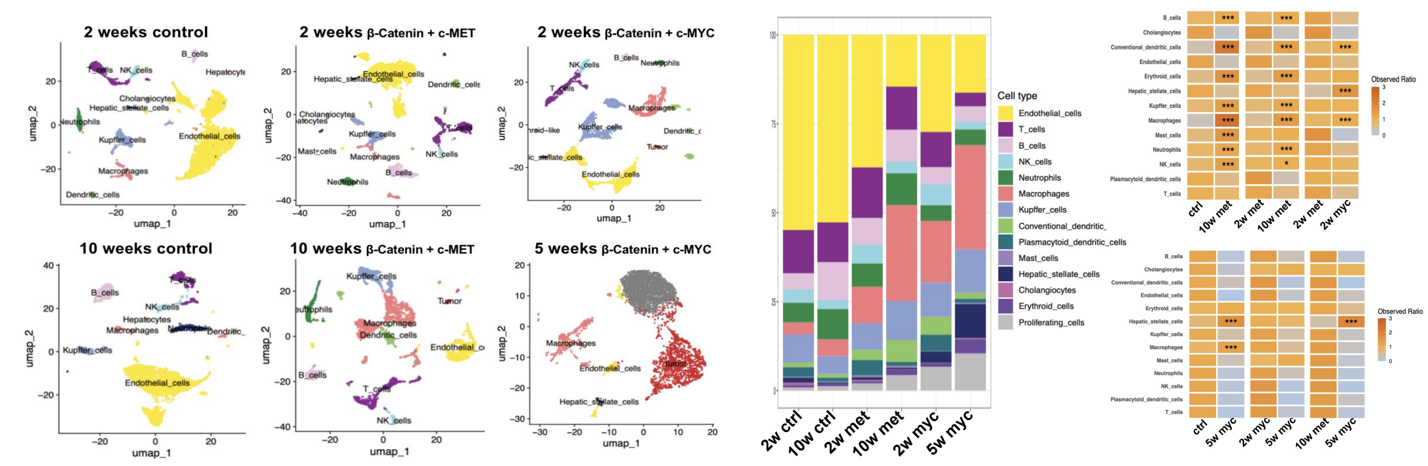
Overview of the study design

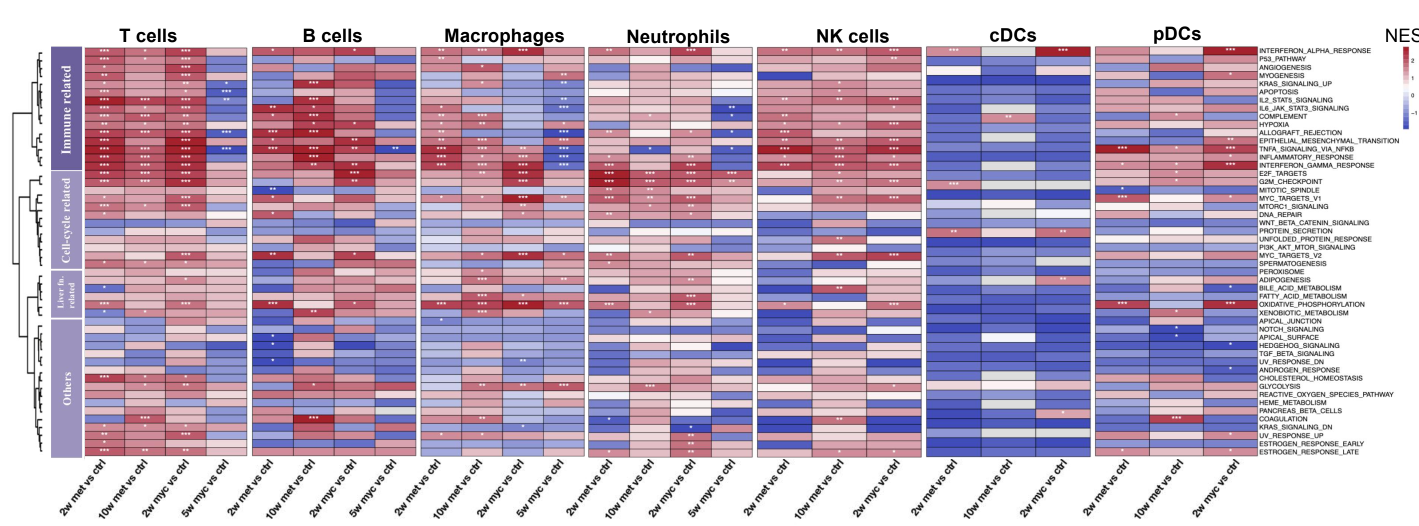


Tumor/non-tumor cell annotation framework based on multiple lines of evidence



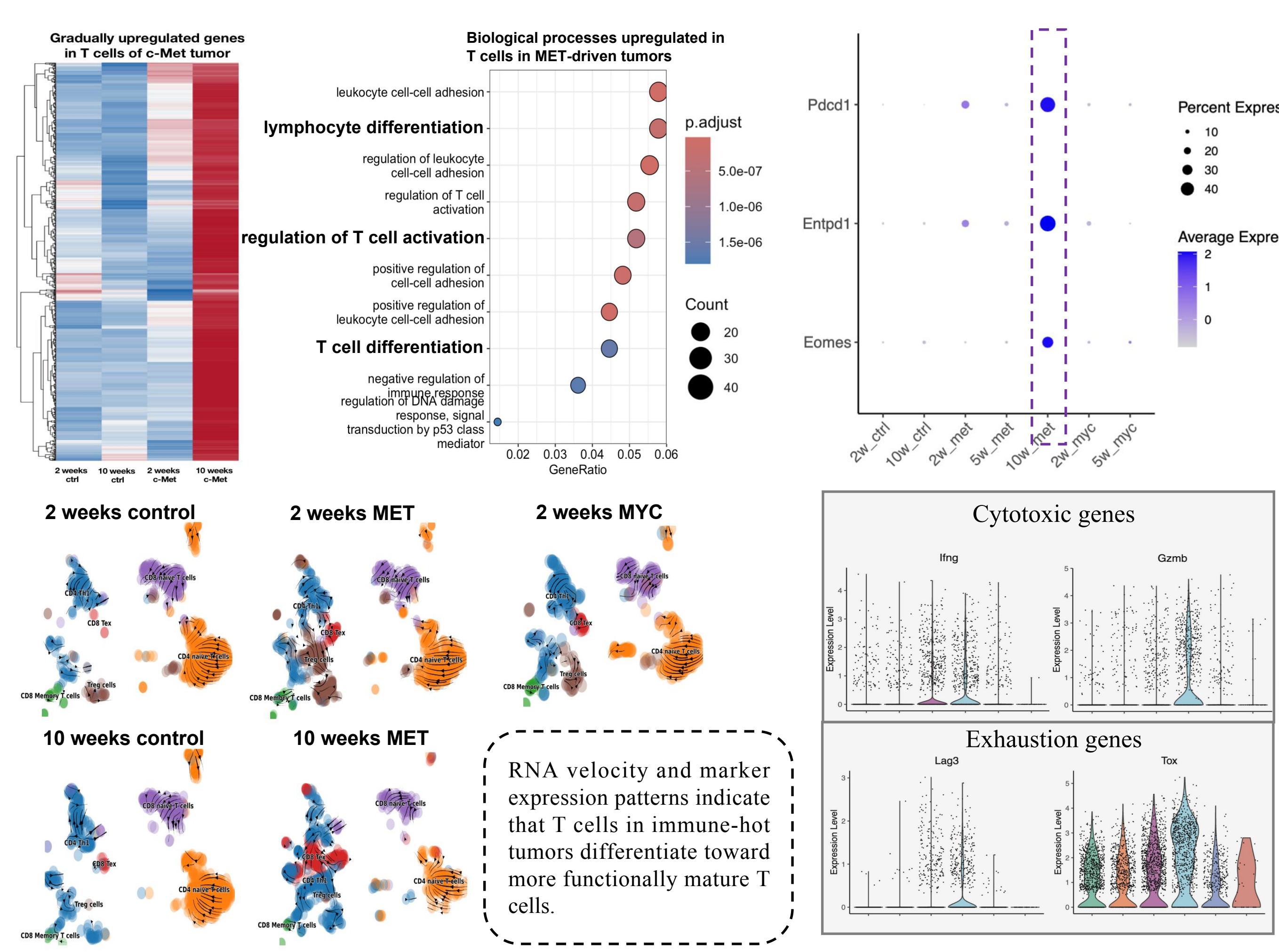
β-catenin + c-Myc and β-catenin + c-Met tumors exhibit cold and hot phenotypes

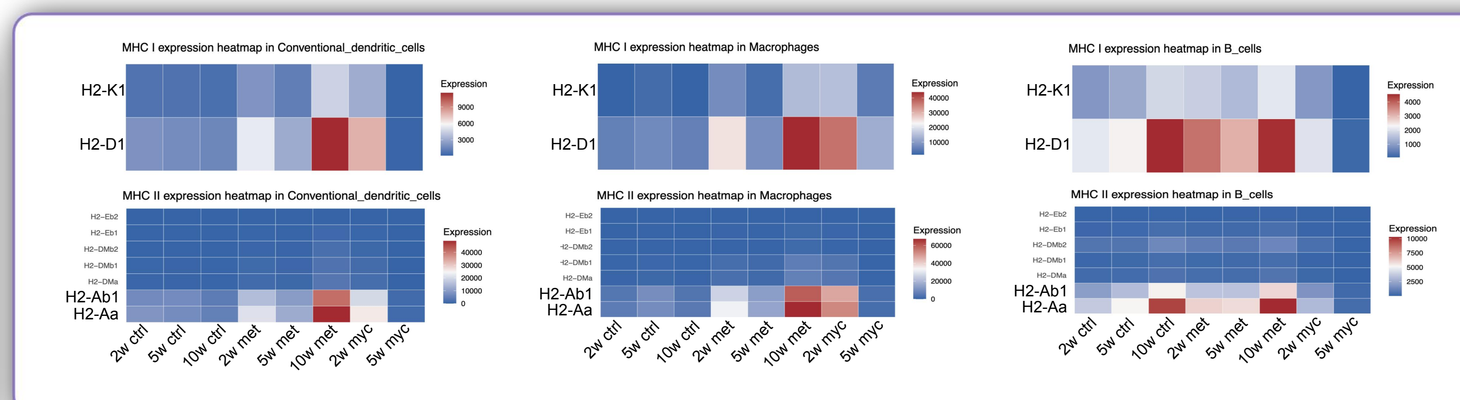




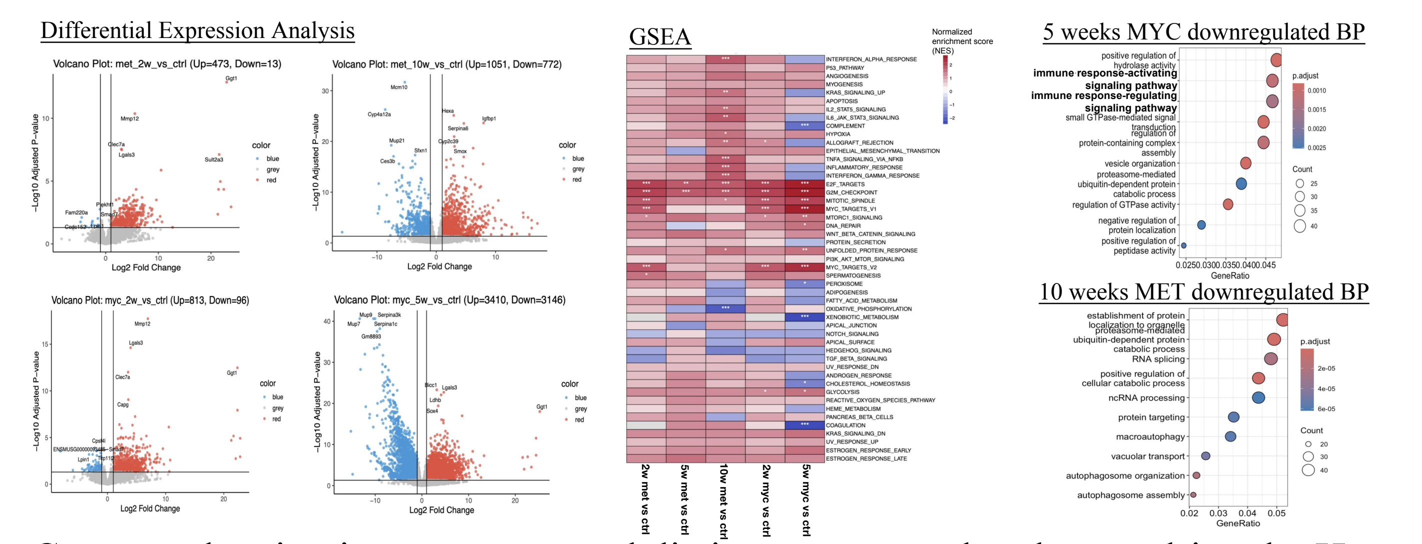
MET-driven tumors show an immune-inflammatory phenotype, with significant increases in various immune cell populations and enhanced immune function, whereas MYC-driven tumors show a decrease in the proportion of lymphocytes.

MET promotes T cell engagement and antigen presentation, MYC suppresses both

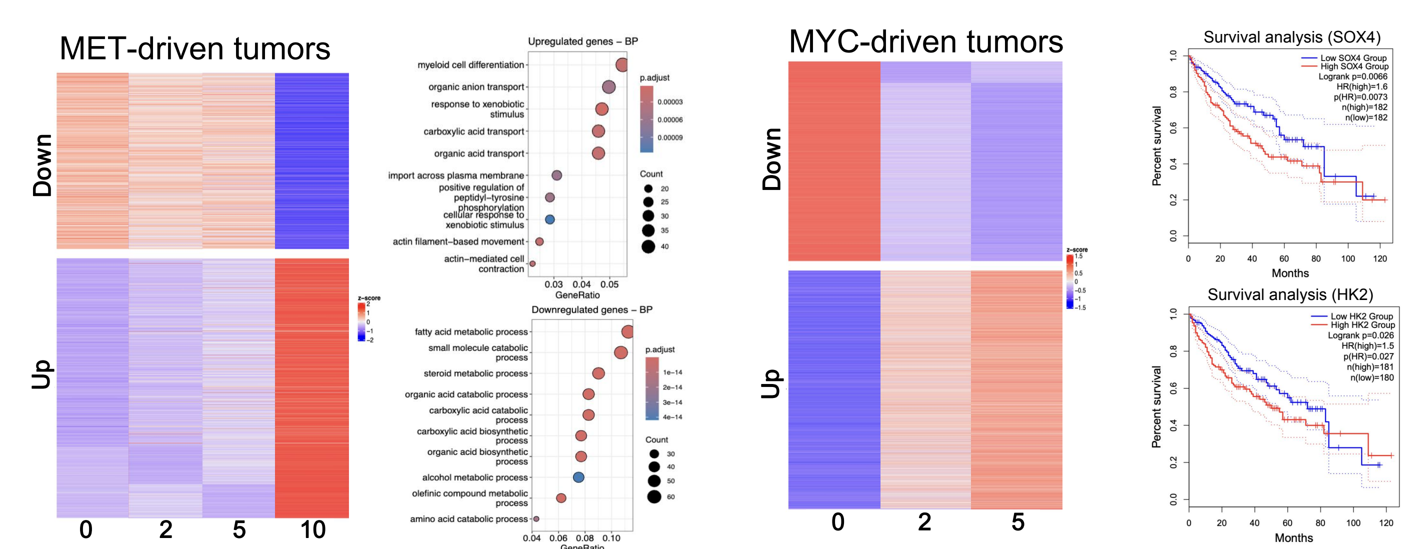




Intrinsic immune characteristics of tumor cells revealed by bulk RNA-seq data



Conserved activation programs and distinct suppressed pathways drive the Hot and Cold tumor phenotypes during tumor progression.



MET-driven tumors exhibit persistent immune activation accompanied by hepatic metabolic suppression. SOX4 and HK2 was continuously upregulated in MYC-driven tumors, consistent with prior studies linking their expression to poor prognosis in HCC.

Conclusion

Our research results indicate that a single oncogenic alteration can profoundly reshape the tumor microenvironment through cell type-specific regulation of antigen presentation and T cell recruitment. Importantly, our findings suggest that SOX4 and HK2 may serve as candidate biomarkers for immune evasion and poor prognosis.

Acknowlegement

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